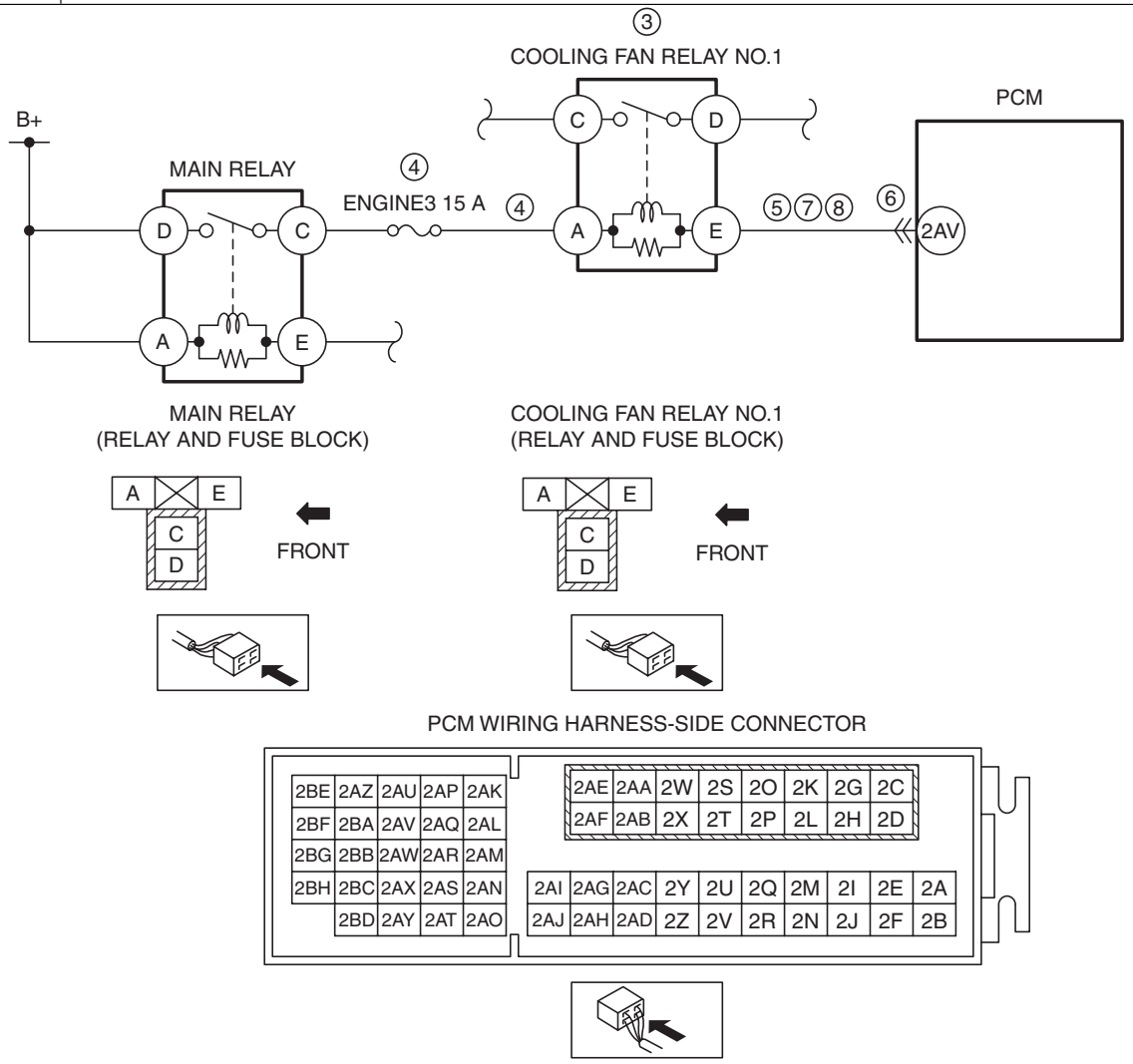
DTC P0482:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0482:00	Cooling fan relay No.1 control circuit malfunction
DETECTION CONDITION	• The cooling fan relay No.1 control voltage at the PCM terminal 2AV is low or cooling fan relay No.1 control current is high for a continuous **5 s** relative to the PCM control signal. **Diagnostic support note** • This is a continuous monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Cooling fan relay No.1 malfunction • Short to ground or open circuit in cooling fan relay No.1 power supply circuit — Short to ground in wiring harness between ENGINE3 15 A fuse and cooling fan relay No.1 terminal A — ENGINE3 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and cooling fan relay No.1 terminal A • Short to ground in wiring harness between cooling fan relay No.1 terminal E and PCM terminal 2AV • PCM connector or terminals malfunction • Short to power supply in wiring harness between cooling fan relay No.1 terminal E and PCM terminal 2AV • Open circuit in wiring harness between cooling fan relay No.1 terminal E and PCM terminal 2AV • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT COOLING FAN RELAY NO.1** • Switch the ignition off. • Remove the cooling fan relay No.1. (See RELAY LOCATION.) • Inspect the cooling fan relay No.1. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the cooling fan relay No.1, then go to Step 9.
		No	Go to the next step.
4	**INSPECT COOLING FAN RELAY NO.1 POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Cooling fan relay No.1 is removed. • Switch the ignition ON (engine off). • Measure the voltage at the cooling fan relay No.1 terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE3 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE3 15 A fuse and cooling fan relay No.1 terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and cooling fan relay No.1 terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

STEP	INSPECTION		ACTION
5	**INSPECT COOLING FAN RELAY NO.1 SIGNAL CIRCUIT FOR SHORT TO GROUND** • Cooling fan relay No.1 is removed. • Switch the ignition off. • Inspect for continuity between cooling fan relay No.1 terminal E (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between cooling fan relay No.1 terminal E and PCM terminal 2AV. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to Step 9.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
7	**INSPECT COOLING FAN RELAY NO.1 SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Cooling fan relay No.1 is removed. • Verify that the PCM connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the cooling fan relay No.1 terminal E (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between cooling fan relay No.1 terminal E and PCM terminal 2AV. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 9.
8	**INSPECT COOLING FAN RELAY NO.1 SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Cooling fan relay No.1 is removed. • Verify that the PCM connector is disconnected. • Switch the ignition off. • Inspect for continuity between cooling fan relay No.1 terminal E (wiring harness-side) and PCM terminal 2AV (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between cooling fan relay No.1 terminal E and PCM terminal 2AV. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION		ACTION
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.